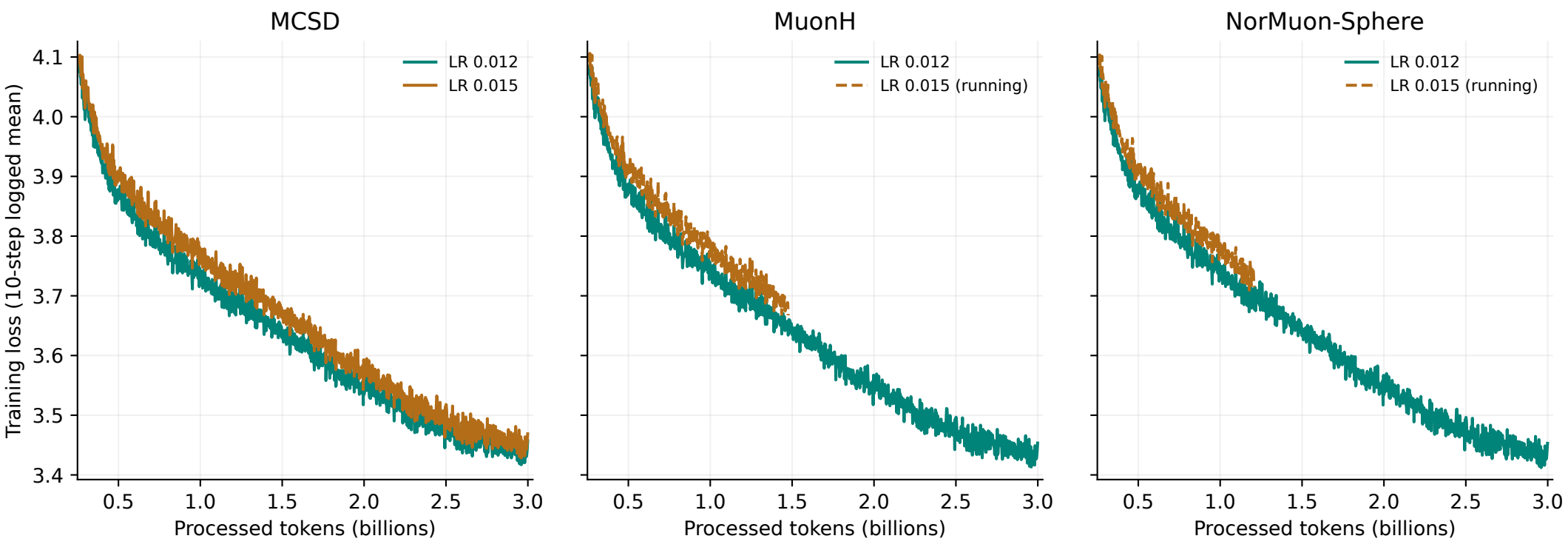

Width 256 | beta=0.95 Nesterov | 3B schedule | Training
MCSD
MuonH
NorMuon-Sphere
LR 0.012
LR 0.015
LR 0.015 (running)
Training loss (10-step logged mean)
4.1
4.0
3.9
3.8
3.7
3.6
3.5
3.4
0.5
1.0
1.5
2.0
2.5
3.0
Processed tokens (billions)
Dashed curves stop at the snapshot; running cells have no final result. No LR=0.010 training trace is included.